Euclid's Elements — Book I
Proposition 15

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Formal Reconstruction — Technical Documentation

If two straight lines cut one another, they make the vertical angles equal to one another.

1 Introduction

Euclid's Proposition I.15 establishes the equality of vertical angles formed by two intersecting straight lines.

If two straight lines intersect at a point, each pair of opposite angles is equal. In modern terminology, these are the vertical-angle relations.

Euclid proves the proposition by applying Proposition I.13 to two adjacent linear pairs and subtracting a common angle from two sums that are each equal to two right angles.

The Hilbert reconstruction is considerably shorter. The corresponding vertical-angle theorem has already been derived in the Hilbert layer as

```
hilbert_vertical_angles
```

from Hilbert's theory of adjacent angles. Proposition I.15 therefore becomes an application of an existing synthetic theorem rather than a new geometric derivation.

This proposition provides another example in which a classical Euclidean proof is compressed by the intermediate Hilbert interface: the mathematical content required by Euclid has already been isolated as a reusable theorem.

2 The Classical Configuration

Let the straight lines AB and CD intersect at the point E, with

$$A - E - B$$

and

$$C - E - D.$$

The intersection determines four angles around E.

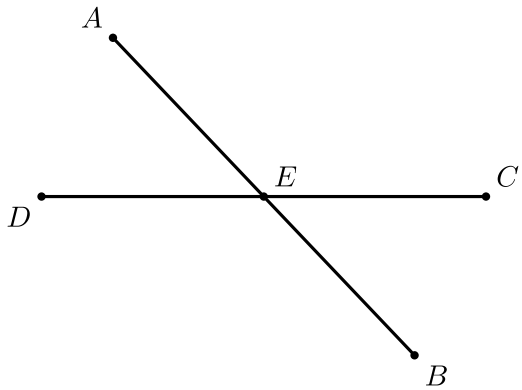


Figure 1: Classical configuration for Proposition I.15. The betweenness relations $A - E - B$ and $D - E - C$ encode the two pairs of opposite rays.

The opposite pairs are

$$\angle AEC \qquad \text{and} \qquad \angle BED,$$

and

$$\angle CEB \qquad \text{and} \qquad \angle DEA.$$

Proposition I.15 states that each pair is equal.

Only one geometric diagram is needed. The two conclusions are not two different configurations: they are the two opposite pairs determined by the same crossing. Repeating the intersection would add no mathematical information.

The betweenness relations

$$A - E - B, \qquad C - E - D$$

express synthetically that EA and EB are opposite rays and that EC and ED are opposite rays. Thus the classical configuration is already naturally adapted to Hilbert's order geometry.

3 Statement

Euclid's Proposition I.15 states:

> If two straight lines cut one another, they make the vertical angles equal to one another.

For the configuration

$$A - E - B, \qquad C - E - D,$$

the proposition gives the two vertical-angle congruences

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

In the Hilbert reconstruction we assume, in addition, that the two lines are genuinely distinct. This is expressed by

$$\neg\,\mathrm{Collinear}(A, E, C).$$

The formal statement is therefore

$$\begin{array}{l} A - E - B, \\ C - E - D, \\ \neg\,\mathrm{Collinear}(A, E, C) \end{array} \qquad \Longrightarrow \qquad \begin{cases} \angle AEC \cong \angle BED, \\ \angle CEB \cong \angle DEA. \end{cases}$$

The noncollinearity hypothesis is the synthetic expression of a proper intersection of two distinct straight lines. It also guarantees that the angles occurring in the proposition are nondegenerate.

Thus the Lean formulation makes explicit a condition that is implicit in Euclid's phrase "two straight lines cut one another." Strict betweenness specifies the two pairs of opposite rays, while noncollinearity excludes the case in which all five points lie on a single straight line.

4 Classical Proof

Euclid first proves the equality of one pair of vertical angles.

Since the straight line AE stands upon the straight line CD, Proposition I.13 gives

$$\angle CEA + \angle AED$$

equal to two right angles.

Again, since the straight line DE stands upon the straight line AB, Proposition I.13 gives

$$\angle AED + \angle DEB$$

equal to two right angles.

Therefore

$$\angle CEA + \angle AED = \angle AED + \angle DEB.$$

Subtracting the common angle $\angle AED$ from both sides, by Common Notion 3, gives

$$\angle CEA = \angle DEB.$$

Thus one pair of vertical angles is equal.

In the same way,

$$\angle CEB = \angle AED.$$

Therefore, when two straight lines cut one another, they make the vertical angles equal to one another.

Wilkins's diagram is especially effective here because the geometry is entirely local at E. The proof needs only the four rays and two linear pairs. The colored sectors in Wilkins's note help distinguish the angles, but no additional construction is involved, so the reconstruction keeps the diagram unshaded and records the equalities in the text.

5 Proof Architecture of the Reconstruction

The formal reconstruction and Euclid's original proof establish the same geometric fact at different levels of the theory.

```
Euclid
------
A-E-B and C-E-D
      |
      +-- CEA + AED = 2R   (I.13)
      |
      +-- DEB + AED = 2R   (I.13)
      |
      v
CEA + AED = DEB + AED
      |
Common Notion 3
      |
      v
CEA = DEB

same argument
      |
      v
CEB = AED


Hilbert / Lean
```

```
--------------
A-E-B, C-E-D, noncollinearity
      |
hilbert_vertical_angles
      |
      v
AEC ~= BED
      |
reorder the same crossing
      |
transport noncollinearity
      |
hilbert_vertical_angles
      |
      v
CEB ~= DEA
```

The formal proof therefore introduces no new geometric point and no new diagrammatic case. Its extra work is purely structural: the second pair is obtained by reorienting the same intersection and proving that the reordered triple is still noncollinear.

Euclid derives the vertical-angle theorem from Proposition I.13. Each of two adjacent angle pairs is equal to two right angles. Since the two sums contain a common angle, Common Notion 3 allows that angle to be subtracted, leaving the two vertical angles equal.

Thus the classical dependency is

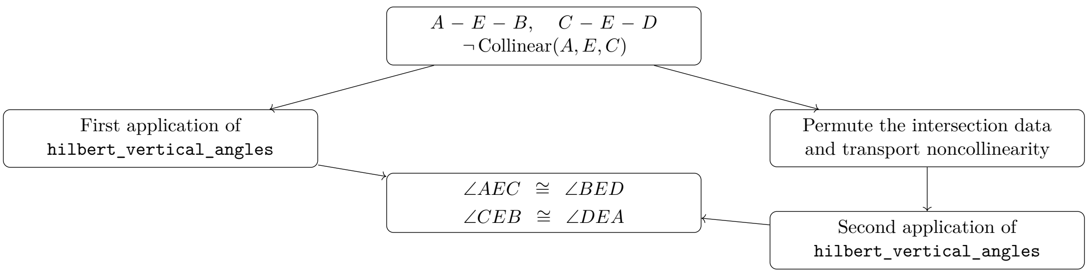
$$\text{I.13} \longrightarrow \text{two linear-pair equalities} \longrightarrow \text{Common Notion 3} \longrightarrow \text{I.15.}$$

The Hilbert development has already isolated the corresponding geometric argument in the theorem

```
hilbert_vertical_angles
```

Consequently, Proposition I.15 does not require a new reconstruction of angle addition or subtraction. The first pair of vertical angles follows by a direct application of this theorem.

The second pair is obtained by viewing the same intersection with the roles of the two straight lines exchanged. The only additional work is to transport the required noncollinearity hypothesis to the reordered triple of points.



Schematically, the two proof architectures are therefore

$$\begin{aligned}\text{Euclid} &: \text{I.13} \to \text{angle sums} \to \text{C.N.3} \to \text{vertical angles},\\ \text{Hilbert reconstruction} &: \text{intersection data} \to \texttt{hilbert\_vertical\_angles} \to \text{vertical angles}.\end{aligned}$$

The difference is architectural rather than mathematical. The Hilbert layer has already packaged the vertical-angle argument as a reusable synthetic theorem, so the Euclidean proposition becomes a thin interface theorem over previously established geometry.

6 Hilbert Reconstruction

The Hilbert reconstruction starts directly from the synthetic data of the intersection:

$$A - E - B, \qquad C - E - D,$$

together with

$$\neg\,\mathrm{Collinear}(A, E, C).$$

Unlike Euclid's proof, no explicit notion of a sum of angles equal to two right angles is required. The relevant geometric content has already been established in the Hilbert layer as the vertical-angle theorem.

6.1 The First Pair of Vertical Angles

The theorem

```
hilbert_vertical_angles
```

has precisely the form needed for the first pair.

Applied to

$$A - E - B, \qquad C - E - D, \qquad \neg\,\mathrm{Collinear}(A, E, C),$$

it gives directly

$$\angle AEC \cong \angle BED.$$

Thus the first conclusion of Proposition I.15 requires no auxiliary construction.

6.2 The Second Pair of Vertical Angles

For the second pair we view the same intersection with the two straight lines exchanged.

From

$$A - E - B$$

the symmetry of strict betweenness gives

$$B - E - A.$$

We also require the noncollinearity condition

$$\neg\,\mathrm{Collinear}(C, E, B).$$

Suppose, for contradiction, that

$$\mathrm{Collinear}(C, E, B).$$

Since $A - E - B$, the points A, E, B are collinear. Permuting the first collinearity relation and using transitivity of collinearity through the distinct points E and B would then imply

$$\mathrm{Collinear}(A, E, C),$$

contrary to the original hypothesis.

Hence

$$\neg\,\mathrm{Collinear}(C, E, B).$$

We may therefore apply `hilbert_vertical_angles` a second time, now to

$$C - E - D, \qquad B - E - A.$$

This yields

$$\angle CEB \cong \angle DEA.$$

Combining the two applications gives the complete conclusion

$$\angle AEC \cong \angle BED, \qquad \angle CEB \cong \angle DEA.$$

The Hilbert reconstruction is therefore substantially shorter than Euclid's original argument. The classical use of Proposition I.13 and Common Notion 3 has already been absorbed into the previously proved vertical-angle theorem; the only additional formal work is the permutation of the intersection data required for its second application.

7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_15
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D E : Geo.Point)
(hAEB : Geo.Between A E B)
(hCED : Geo.Between C E D)
(hAEC : not PrimCollinear Geo A E C) :
Geo.AngleCongruent A E C B E D /\
Geo.AngleCongruent C E B D E A := by
```

The conclusion records both pairs of vertical angles:

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

The first pair is obtained directly:

```
have hFirst :
Geo.AngleCongruent A E C B E D :=
hilbert_vertical_angles
Geo A E C B D
hAEB
hCED
hAEC
```

For the second pair, the first line is read in the opposite direction:

```
have hBEC : Geo.Between B E A :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.2
```

The required noncollinearity condition is then derived by contradiction. If C, E, B were collinear, cyclic permutation gives E, B, C collinear. Since A, E, B are already collinear and $E \neq B$, transitivity would imply that A, E, C are collinear, contradicting `hAEC`.

In Lean this is expressed by

```
have hCEB :
not PrimCollinear Geo C E B := by
intro h
```

```
have hEBC :
PrimCollinear Geo E B C :=
PrimCollinearCycle Geo C E B h

have hAEBcol :
PrimCollinear Geo A E B :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.1

have hEB : E != B :=
(HilbertOrder.between_incidence
A E B hAEB).2.1

have hAECcol :
PrimCollinear Geo A E C :=
hilbert_primCollinear_trans
Geo A E B C
hEB
hAEBcol
hEBC

exact hAEC hAECcol
```

The second pair of vertical angles is now another direct application:

```
have hSecond :
Geo.AngleCongruent C E B D E A :=
hilbert_vertical_angles
Geo C E B D A
hCED
hBEC
hCEB

exact <hFirst, hSecond>
```

Thus the formal dependency structure is

`hAEB`, `hCED`, `hAEC`

↓

`hilbert_vertical_angles`
first orientation

↓

Betweenness symmetry and
collinearity transport

↓

`hilbert_vertical_angles`
second orientation

↓

$\angle AEC \cong \angle BED$
$\angle CEB \cong \angle DEA$

No new geometric axiom or auxiliary geometric construction is introduced in the formalization. Proposition I.15 is obtained entirely from the existing Hilbert incidence, order, collinearity, and vertical-angle theory.

8 Relationship with Earlier Propositions

8.1 Proposition I.13

Proposition I.13 establishes the supplementary-angle configuration created when a straight line stands on another straight line. That result belongs to the same local geometry of intersecting lines that underlies I.15.

The logical roles are different, however. I.13 concerns adjacent angles whose sum forms a straight angle; I.15 concerns opposite angles and proves their congruence.

8.2 Proposition I.14

Proposition I.14 gives a converse criterion for recognizing a straight line from a supplementary configuration. Together, I.13 and I.14 develop the straight-angle infrastructure surrounding an intersection.

Proposition I.15 then extracts a different invariant of the same configuration: opposite, or vertical, angles are congruent.

8.3 Transition to Proposition I.16

Vertical-angle congruence becomes an explicit transport tool in Proposition I.16. The exterior-angle proof naturally produces angles in orientations that do not always match the public Euclidean statement, and I.15 supplies the congruence needed to move between those orientations.

Thus I.15 is not merely the end of the elementary intersecting-lines block. It becomes part of the infrastructure for the strict angle-comparison theory beginning with I.16.

9 Hilbert and Book Zero Components Used

Proposition I.15 is formally short because its main geometric content has already been isolated in the Hilbert layer.

The proof uses one principal theorem together with elementary order and collinearity machinery required to apply it in the two orientations of the intersection.

9.1 `hilbert_vertical_angles`

The central result is

```
hilbert_vertical_angles
```

which states that if

$$C - E - B, \qquad D - E - F,$$

and

$$\neg\,\mathrm{Collinear}(C, E, D),$$

then

$$\angle CED \cong \angle BEF.$$

This theorem is the immediate vertical-angle corollary developed in the Hilbert layer after the theorem on adjacent congruent angles.

In Proposition I.15 it is used twice.

The first application gives

$$\angle AEC \cong \angle BED.$$

The second application, after reversing the first straight line and transporting noncollinearity, gives

$$\angle CEB \cong \angle DEA.$$

Thus essentially all of the geometric content of Euclid's Proposition I.15 is already contained in `hilbert_vertical_angles`.

9.2 `HilbertOrder.between_incidence`

The theorem

```
HilbertOrder.between_incidence
```

extracts the elementary order and incidence consequences of strict betweenness.

From

$$A - E - B$$

the proof obtains both

$$B - E - A$$

and the collinearity relation

$$\text{Collinear}(A, E, B).$$

It also supplies the inequality

$$E \neq B,$$

which is needed for the transitivity of collinearity.

Thus a single betweenness hypothesis provides all of the auxiliary order data required for the second application of the vertical-angle theorem.

9.3 `PrimCollinearCycle`

The permutation theorem

```
PrimCollinearCycle
```

cyclically reorders a collinearity relation.

In the contradiction argument,

$$\text{Collinear}(C, E, B)$$

is transformed into

$$\text{Collinear}(E, B, C).$$

This places the points in exactly the orientation required by the transitivity theorem used in the next step.

9.4 `hilbert_primCollinear_trans`

The theorem

```
hilbert_primCollinear_trans
```

expresses the transitivity of collinearity when two distinct common points determine the same line.

In Proposition I.15 we combine

$$\mathrm{Collinear}(A, E, B)$$

with

$$\mathrm{Collinear}(E, B, C)$$

and the inequality

$$E \neq B.$$

The result is

$$\mathrm{Collinear}(A, E, C),$$

contradicting the original noncollinearity hypothesis.

Hence

$$\neg\,\mathrm{Collinear}(C, E, B),$$

which is precisely the hypothesis needed for the second invocation of `hilbert_vertical_angles`.

The complete formal dependency may therefore be summarized as

$$\begin{gathered} A - E - B, \quad C - E - D, \quad \neg\,\mathrm{Collinear}(A, E, C) \\ \downarrow \\ \texttt{hilbert\_vertical\_angles} \\ \downarrow \\ \angle AEC \cong \angle BED \\ \text{betweenness symmetry} + \text{collinearity transport} \\ \downarrow \\ \neg\,\mathrm{Collinear}(C, E, B) \\ \downarrow \\ \texttt{hilbert\_vertical\_angles} \\ \downarrow \\ \angle CEB \cong \angle DEA. \end{gathered}$$

No genuinely new Book Zero theorem is required for Proposition I.15. The proposition is therefore primarily an interface theorem over the already established Hilbert vertical-angle theory.

10 Formalization Notes

Proposition I.15 is classically almost immediate from the supplementary angle theory, but its formal reconstruction highlights a different issue: orientation.

At an intersection there are two pairs of vertical angles. The geometric theorem is symmetric, while the formal predicates are written with ordered point triples. Consequently the proof must explicitly recover the appropriate collinearities from betweenness, rotate or cycle them into the required orientation, and then invoke the reusable vertical-angle theorem.

The essential pattern is

$$\text{betweenness} \longrightarrow \text{primitive collinearity} \longrightarrow \text{orientation normalization} \longrightarrow \texttt{hilbert\_vertical\_angles} \longrightarrow \text{angl}$$

This is a good example of why the Book Zero and Hilbert interface layers contain many apparently small symmetry and rotation lemmas. On paper, one simply says that two lines intersect. In Lean, the exact ordering of the points on those lines determines which theorem application type-checks.

No new geometric axiom is introduced by I.15. The proposition packages the already established intersection and angle-congruence machinery into the classical Euclidean statement.

11 Dependency Summary

The formal dependency structure is

$$\begin{array}{c}
\text{two intersecting straight lines} \\
\downarrow \\
\text{betweenness hypotheses} \\
\downarrow \\
\texttt{HilbertOrder.between_incidence} \\
\downarrow \\
\text{collinearity rotation / transitivity} \\
\downarrow \\
\texttt{hilbert_vertical_angles} \\
\downarrow \\
\text{first pair of vertical angles congruent} \\
\text{second pair of vertical angles congruent} \\
\downarrow \\
\texttt{euclid_proposition_15}.
\end{array}$$

12 Conclusion

Proposition I.15 completes the elementary theory of angles formed by intersecting straight lines.

Its classical content is the familiar theorem that vertical angles are congruent. The formal reconstruction shows that the principal burden is not angle arithmetic but the management of incidence and orientation: betweenness must be converted into primitive collinearity, and the resulting triples must be placed in exactly the orientation expected by the vertical-angle theorem.

The proposition is also structurally important for what follows. Vertical-angle congruence becomes a reusable transport mechanism in Proposition I.16, where Book I moves from congruence statements to strict angle comparison.

Thus I.15 closes one local geometric block and simultaneously provides infrastructure for the next.